These are notes I'm taking for Ben Hayes's Ergodic Theory class for Fall 2026.

The main texts are Petersen's *Ergodic Theory* and Walters's *An Introduction to Ergodic Theory*, as well as some other

General Motivation and Common Examples

The basic structure of ergodic theory is a standard probability space (X, μ) and, in its most basic form, a measurable map $T : X \to X$ satisfying $T_*\mu = \mu$. Recall that the pushforward measure is defined by

$$T_*\mu(A) = \mu\Big(T^{-1}(A)\Big).$$

We call T a measure-preserving transformation.

Generalizing this, if T is invertible, then we may take a group action $G \curvearrowright (X, \mu)$ by measure-preserving transformations. If T is not invertible, then we may consider a semigroup action $N \curvearrowright (X, \mu)$.

The main items we are interested in are analytical considerations, such as

- asymptotic structure
- connections to unitary representation theory
- isomorphism.

Ergodic theory appears in a number of fields, such as

- unitary representation theory
- analytic and geometric group theory
- functional analysis
- probability theory
- geometry
- analytic number theory.

Background

More aptly, this may be written as "background." Most of these results are covered in a functional analysis class.

Definition: A measurable space (X, Σ) is called *standard Borel* if there is a separable and completely metrizable space (also known as a Polish space) K such that $(K, \mathcal{B}) \cong (X, \Sigma)$ are isomorphic as measurable spaces, where $\mathcal{B}$ denotes the Borel σ-algebra on K.

Remark: In the special case of standard probability spaces, we may take K to be compact. This emerges from the fact we are only concerned with Radon measures.

Example: The following are examples:

- finite or countably infinite sets;
- products of standard Borel spaces;
- $[0, 1) \cong S^1$;
- $(0, 1) \cong \mathbb{R} \cong [0, 1) \times \mathbb{Z}$;
- $\mathbb{R}^{\mathbb{N}}$, where we use the complete distance metric

$$d(t, s) = \sum_{j=1}^{\infty} 2^{-j} \frac{\big|t_j - s_j\big|}{1 + \big|t_j - s_j\big|}$$

to provide the desired isomorphism;

- the set $\mathbb{R} \setminus \mathbb{Q}$ is a G_δ subset of the Polish space $\mathbb{R}$, so it is also a Polish space.

Theorem: If (X, Σ) is standard Borel and B is Borel, then $(B, \Sigma|_B)$ is standard Borel.

Theorem (Riesz Representation Theorem)**:** Let K be a compact Hausdorff space, and $L\colon C(K) \to \mathbb{C}$ a continuous linear functional. Then, there is a unique complex-valued Borel measure such that

$$L(f) = \int f \, d\mu.$$

Furthermore, this measure is a Radon measure.

Remark: From the Riesz Representation Theorem, it can be shown that the following hold:

- L is positive if and only if μ is a positive measure;
- $L(1) = 1$ if and only if μ is a probability measure.

One of the fundamental constructions in ergodic theory, especially when we are concerned with actions by a countable discrete group on a standard probability space, is the projective limit of a (countable) family of standard probability spaces. This is a vital theorem.

Consider a system of standard probability spaces $(X_k, \Sigma_k)_{k=0}^{\infty}$ with connecting maps $\varphi_{k+1}\colon X_{k+1} \to X_k$ satisfying $(\varphi_{k+1})_*\mu_{k+1} = \mu_k$.

Then, the projective limit

$$\begin{aligned} X_\infty &= \varprojlim X_k \\ &= \left\{ (x_k)_k \in \prod_{k=1}^{\infty} X_k : \varphi_k(x_{k+1}) = x_k \right\} \end{aligned}$$

admits a family of projection maps $\varphi_{k,\infty}\colon X_\infty \to X_k$, where we have $\varphi_{k,\infty}((x_n)_n) = x_k$ such that $\varphi_k \circ \varphi_{k+1,\infty} = \varphi_{k,\infty}$.

$$\begin{array}{ccc} & X_\infty & \\ {\scriptstyle \varphi_{k+1,\infty}}\swarrow & & \searrow{\scriptstyle \varphi_{k,\infty}} \\ X_{k+1} & \xrightarrow[\varphi_k]{} & X_k \end{array}$$

The projective limit is universal in the sense that if we have a family of maps $f_{k+1}\colon Y \to X_{k+1}$ satisfying $\varphi_k \circ f_{k+1} = f_k$, then there is a unique map $f\colon Y \to X$ such that $\varphi_{k,\infty} \circ f = f_k$ for all k.

We define the σ-algebra on X_∞ to be

$$\Sigma_\infty = \sigma\left(\bigcup_{k=1}^{\infty} \left\{ \varphi_{k,\infty}^{-1}(E_k) : E_k \in \Sigma_k \right\} \right).$$

Then, the **Kolmogorov Extension Theorem** says the following.

Theorem (Kolmogorov Extension Theorem)**:** There is a unique probability measure μ_∞ on $(X_\infty, \Sigma_\infty)$ satisfying

$$\mu_\infty\left(\varphi_{k,\infty}^{-1}(E)\right) = \mu_k(E)$$

for all $E \in \Sigma_k$.

We will give two proofs: one uses the Carathéodory criterion and is more explicit, while the other uses the Riesz Representation Theorem and is more slick.

Proof via the Riesz Representation Theorem. Since all the maps φ_k are Borel, there are zero-dimensional Polish topologies on each set X_k such that the φ_k are continuous; since the X_k are all probability spaces and

the measures μ_k are regular, it follows that each X_k is compact. In particular, we also have that the maps $\varphi_{k,\infty}$ are continuous.

Using Gelfand duality, it follows that for each X_k, there is an embedding $C(X_k) \to C(X_\infty)$ such that for any $f \in C(X_k)$, we obtain a continuous map $f \circ \varphi_{k,\infty} \colon X_\infty \to \mathbb{C}$. The set

$$A = \bigcup_{k \geq 1} C(X_k),$$

viewing each $C(X_k)$ as a subset of $C(X_\infty)$, is a unital $*$-subalgebra that separates the points of X_∞. We may define the linear functional $L \colon A \to \mathbb{C}$ given by taking

$$L|_{C(X_k)} = \int \cdot \, d\mu_k.$$

Notice that this is a positive linear functional satisfying $L(1) = 1$, so since L is continuous on the unital $*$-subalgebra A, it follows that there is a unique extension to the closure $L_\infty \colon \overline{A} = C(X_\infty) \to \mathbb{C}$ satisfying $L_\infty(1) = 1$. By the Riesz Representation Theorem, there is a unique measure μ_∞ such that

$$L_\infty(f) = \int f \, d\mu_\infty$$

and satisfies

$$L_\infty|_{C(X_k)} = \int \cdot \, d\mu_k.$$

□

Proof via Carathéodory Criterion. Without loss of generality, we may assume that the X_k are compact and the φ_{k+1} are Borel. Consider the algebra

$$\mathcal{A} = \bigcup_{k \geq 1} \left\{ \varphi_{k,\infty}^{-1}(E) : E \in \Sigma_k \right\}.$$

Notice that this is the algebra of subsets that generates Σ_∞. Furthermore, there is a unique set-map $\mu_\infty \colon \mathcal{A} \to [0,1]$ taking $\mu_\infty\left(\varphi_{k,\infty}^{-1}(E)\right) = \mu_k(E)$ for all $E \in \Sigma_k$. Thus, it suffices to show that μ_∞ is a premeasure.

It suffices to show that for all decreasing sequences $(F_n)_n$ with $\mu_\infty(F_n) \geq \varepsilon > 0$, then the intersection of all the F_n is nonempty. This follows from the fact that if one has an ascending sequence $E_n \nearrow E$, then $(E \setminus E_n) \searrow \emptyset$, so if one takes the Carathéodory criterion and takes the contrapositive of this reversal, we obtain the desired result.

Let

$$F_k = \varphi_{n_k,\infty}^{-1}\left(E_{n_k}\right).$$

If n_k does not diverge to ∞, then there exists some subsequence n_{k_j} that is constant equal to n, and F_{k_j} all have $\varphi_{n,\infty}\left(F_{k_j}\right) = E_{n_{k_j}}$. Since μ_n is a measure, we are done in this case, following from the fact that all the F_n past F_{k_j} lie in X_n. □

The most important case is when the projective limit is the infinite product measure. If (X_n, μ_n) is a countable family of standard probability spaces, then the product measure $\bigotimes_{n=1}^{\infty} \mu_n$ is a probability measure.

Canonical Examples

Example (Bernoulli Actions)**:** Let (X, μ) be a standard probability space, and let G be any countable group. Then, there is a unique probability measure on the product space $(X, \mu)^G$, and we have that $G \curvearrowright (X, \mu)^G$ in a probability measure-preserving manner by translation — that is, $h \cdot \left(x_g\right)_{g \in G} = \left(x_{h^{-1}g}\right)_{g \in G}$. This follows from the fact that G is probability measure-preserving on all the Borel cylinder sets

$$\bigcup_{\substack{F \subseteq G \\ F \text{ finite}}} \left\{\pi_F^{-1}(E) : E \subseteq X^F\right\},$$

so we may apply the monotone class lemma (that no one remembers from measure theory) to see that it applies for all the Borel sets of $(X, \mu)^G$.

Example: Suppose G is a compact Hausdorff topological group. Then, there is a unique (regular) Borel probability measure μ, known as the Haar (probability) measure that is invariant under left translation. If $\pi : \Gamma \to G$ is a group homomorphism of a discrete group Γ on G, then the action $\gamma \cdot g = \pi(\gamma)g$ yields a probability measure-preserving action of Γ on G.

For instance, this yields actions $\mathbb{Z} \curvearrowright S^1$ given by rotations — that is, $e^{2\pi i\theta} \mapsto e^{2\pi i(\theta+\alpha)}$.

In general, if G is a compact group, and $\phi : \Gamma \to \mathrm{Aut}(G)$ is an action of a discrete group Γ that preserves the Haar measure, we call such an action an *algebraic action*.

Example: If M is a finite volume Riemannian manifold with volume form vol, and $\phi \in \mathrm{Diff}(M)$ is a diffeomorphism satisfying $\det(\phi'(p)^*\phi'(p)) = 1$ for all $p \in M$, then the pushforward measure satisfies $\phi_* \mathrm{vol} = \mathrm{vol}$.

Group actions $\Gamma \to \mathrm{Diff}(M)$ appear in geodesic and horocycle flow.

The Ergodic Theorems

We will now discuss the mean and pointwise ergodic theorems.

Mean Ergodic Theorem

Proposition: Let X be a Banach space. Let $(T_n)_n \in \mathbb{B}(X)$ be a uniformly operator-norm bounded. Then, for all $S \in \mathbb{B}(X)$, the sets

$$V = \{x \in X : T_n x \text{ converges}\}$$
$$W = \{x \in X : \|T_n x - Sx\| \to 0\}$$

are norm-closed subspaces of X.

Theorem (Mean Ergodic Theorem)**:** Let H be a Hilbert space, and let $T \in \mathbb{B}(H)$ be such that $\|T\| \leq 1$. Define

$$V = \{\xi \in H : T\xi = \xi\},$$

and let $P = \mathrm{proj}_V$. Then,

$$\left(\frac{1}{n}\sum_{j=0}^{n-1} T^j\right) \xrightarrow{\text{SOT}} P.$$

Proof. Let $K_1 = \mathrm{im}(1-T)$ and $K_2 = \ker(1-T)$. We claim that K_1 and K_2 are orthogonal to each other and satisfy $\overline{K_1} + K_2 = H$. For the first point, we have for all $v \in K_1$ and $w \in K_2$, we write $v = (1-T)\xi$, and have

$$\langle (1-T)\xi, w\rangle = \langle \xi, w\rangle - \langle \xi, T^* w\rangle$$

$$\begin{aligned}\|T^*w - w\|^2 &= \|T^*w\|^2 - 2\,\mathrm{Re}(\langle w, Tw\rangle) + \|w\|^2 \\ &= \|T^*w\|^2 - \|w\|^2 \\ &\leq 0,\end{aligned}$$

where we note that $\|T^*\| = \|T\|$. Therefore, $T^*w = w$, so we have that $K_2 \subseteq K_1^\perp$.

Now, suppose $\xi \in K_1^\perp$. Then,

$$\|(1-T)\xi\|^2 = \|T\xi\|^2 - 2\,\mathrm{Re}(\langle T\xi, \xi\rangle) + \|\xi\|^2,$$

with

$$\begin{aligned}\langle T\xi, \xi\rangle &= \langle (T-1)\xi, \xi\rangle + \|\xi\|^2 \\ &= \|\xi\|^2\end{aligned}$$

since $\xi \in K_1^\perp$. Therefore,

$$\begin{aligned}\|(1-T)\xi\|^2 &\leq \|\xi\|^2 - \|\xi\|^2 \\ &= 0\end{aligned}$$

since $\|T\| \leq 1$. Therefore, $K_2 = K_1^\perp$.

Therefore, it suffices to show that

$$\frac{1}{n}\sum_{j=0}^{n-1} T^j\xi \to P\xi$$

for all $\xi \in K_1 \cup K_2$. For each $\xi \in K_2$, we obtain the desired result since both sides are equal to ξ.

For each $\xi \in K_1$, we have that $P\xi = 0$. If $\xi = (1-T)\eta$, then

$$\frac{1}{n}\sum_{j=0}^{n-1} T^j\xi = \frac{1}{n}(1-T^n)\eta,$$

so that

$$\begin{aligned}\left\|\frac{1}{n}\sum_{j=0}^{n-1} T^j\xi\right\| &\leq \frac{2}{n}\|\eta\| \\ &\xrightarrow{n\to\infty} 0.\end{aligned}$$

□

Pointwise Ergodic Theorem

Now, consider a probability space (X, μ) and a measure-preserving map $T\colon X \to X$. Then, the operator $V_T \in \mathbb{B}\big(L^2(X,\mu)\big)$ is given by $V_T f = f \circ T$ is an isometry.

If $f \in L^\infty(X,\mu)$, and $P = \mathrm{proj}(\mathrm{fix}(V_T))$, then we have $\|Pf\|_{L^\infty} \leq \|f\|_{L^\infty}$. This follows from the fact that we may find a subsequence such that

$$\frac{1}{n_k}\sum_{j=0}^{n_k-1} f \circ T^j \to Pf$$

pointwise almost everywhere. Therefore, we have

$$\begin{aligned}\|Pf\|_{L^\infty} &\leq \liminf_{k\to\infty} \frac{1}{n_k} \left\| \sum_{j=0}^{n_k-1} f \circ T^j \right\|_{L^\infty} \\ &\leq \|f\|_{L^\infty}.\end{aligned}$$

A similar result holds on L^p for general p. We discuss this below.

Corollary: Suppose (X, μ) is a probability space, and $T : X \to X$ is a probability measure-preserving action. Then, for all $f \in L^p(X, \mu)$ with $1 \leq p < \infty$, there exists $\widetilde{f} \in L^p(X, \mu)$ such that

$$\left\| \widetilde{f} - \frac{1}{n} \sum_{j=0}^{n-1} f \circ T^j \right\|_{L^p} \xrightarrow{n\to\infty} 0.$$

The element $\widetilde{f}$ satisfies:

(i) $\widetilde{f} \circ T = \widetilde{f}$ almost everywhere;

(ii) $\displaystyle\int \widetilde{f} g \, d\mu = \int f g \, d\mu$ for all $g \in L^q(X, \mu)$ satisfying $g \circ T = g$ almost everywhere, where q is the conjugate exponent for p;

(iii) $\left\|\widetilde{f}\right\|_{L^p} \leq \|f\|_{L^p}$.

Proof. In the following proof, we will still be concerned with the operator P defined as

$$\begin{aligned}P &= \text{proj}(\text{fix}(V_T)) \\ &= \text{proj}\Big(\Big\{f \in L^2(X, \mu) : V_T f = f\Big\}\Big).\end{aligned}$$

(i) Let $V_T \in \mathbb{B}(L^p(X, \mu))$ be defined by $V_T f = f \circ T$. Then,

$$\left\| \frac{1}{n} \sum_{j=0}^{n-1} V_T^j \right\| \leq 1.$$

We have that

$$W = \left\{ f \in L^p(X, \mu) : \frac{1}{n} \sum_{j=0}^{n-1} V_T^j f \text{ converges} \right\}$$

is a norm-closed subspace. It suffices to show that $L^\infty(X, \mu) \subseteq W$, as then the L^p-closure of $L^\infty(X, \mu)$ is contained in W, but that is equal to $L^p(X, \mu)$.

Fix $f \in L^\infty(X, \mu)$.

If $1 \leq p \leq 2$, we have that

$$\begin{aligned}\left\| \frac{1}{n} \sum_{j=0}^{n-1} V_T^j f - Pf \right\|_{L^p} &\leq \left\| \frac{1}{n} \sum_{j=0}^{n-1} V_T^j f - Pf \right\|_{L^2} \\ &\to 0.\end{aligned}$$

If $p \geq 2$, then

$$\begin{aligned}\left\| \frac{1}{n} \sum_{j=0}^{n-1} V_T^j f - Pf \right\|_{L^p}^p &\leq 2\|f\|_{L^\infty}^{p-2} \left\| \frac{1}{n} \sum_{j=0}^{n-1} V_T^j f - Pf \right\|_{L^2}^2 \\ &\to 0.\end{aligned}$$

(ii) We have that $(fg)\circ T^j = \big(f\circ T^j\big)g$, so that

$$\left\|\widetilde{fg} - \frac{1}{n}\sum_{j=0}^{n-1}\big(f\circ T^j\big)g\right\|_{L^1} \leq \|g\|_q \left\|\widetilde{f} - \frac{1}{n}\sum_{j=0}^{n-1} V_T^j f\right\|_{L^p}.$$

In particular, we may write $\widetilde{fg}$ using the asymptotic form for $\widetilde{f}$. Therefore, we have

$$\begin{aligned}\int \widetilde{fg}\,d\mu &= \lim_{n\to\infty}\frac{1}{n}\sum_{j=0}^{n-1}\int f\Big(T^j(x)\Big)g(x)\,d\mu(x)\\ &= \lim_{n\to\infty}\frac{1}{n}\sum_{j=0}^{n-1}\int (fg)\Big(T^j(x)\Big)\,d\mu(x)\\ &= \int fg\,d\mu.\end{aligned}$$

(iii) We have that

$$\begin{aligned}\left\|\widetilde{f}\right\|_{L^p} &= \lim_{n\to\infty}\frac{1}{n}\left\|\sum_{j=0}^{n-1} f\circ T^j\right\|_{L^p}\\ &\leq \lim_{n\to\infty}\frac{1}{n}\sum_{j=0}^{n-1}\widetilde{f}\circ T^j_{L^p}\\ &= \|f\|_{L^p}.\end{aligned}$$

□

The next result we will be concerned with is the Pointwise Ergodic Theorem. To prove it, we must prove a version of this theorem known as the Maximal Ergodic Theorem. This concerns a maximal function defined by

$$f^*(x) = \sup_n \frac{1}{n}\sum_{j=0}^{n-1} f\circ T^j(x).$$

Theorem (Maximal Ergodic Theorem)**:** If $f\in L^1(X,\mu)$, then

$$\int_{\{x:f^*(x)>0\}} f\,d\mu \geq 0.$$

Proof. The set on which f^* is strictly greater than zero is the disjoint union of the sets

$$\begin{aligned}B_1 &= \{x : f(x)\geq 0\}\\ &\vdots\\ B_n &= \Big\{x : f(x)\leq 0, f(x)+f\circ T(x)\leq 0,\ldots, f(x)+\cdots+f\circ T^{n-2}(x)\leq 0, f(x)+\cdots+f\circ T^{n-1}(x)>0\Big\}.\end{aligned}$$

If we are able to show that

$$\int_{B_1\cup\cdots\cup B_n} f\,d\mu \geq 0$$

holds, then dominated convergence yields the desired result.

In order to obtain this, we will break up the set $B_1\cup\cdots\cup B_n$ into a family of disjoint pieces on which the

integral of f is strictly positive. We will view them as towers of the form $B'_k \cup T\left(B'_k\right) \cup \cdots \cup T^{k-1}\left(B'_k\right)$.

First, we observe that $T^k(B_n) \subseteq B_1 \cup \cdots \cup B_{n-k}$ for each $k = 1, \ldots, n-1$, since if $x \in B_n$, then we have

$$0 \geq f(x) + f \circ T(x) + \cdots + f \circ T^{k-1}(x)$$

while

$$0 < f(x) + f \circ T(x) + \cdots + f \circ T^{k-1}(x) + f \circ T^k(x) + \cdots + f \circ T^{n-1}(x).$$

In particular, this means that

$$\begin{aligned} 0 &< f \circ T^k(x) + \cdots + f \circ T^{n-1}(x) \\ &= f\left(T^k(x)\right) + f \circ T\left(T^k(x)\right) + \cdots + f \circ T^{n-k-1}\left(T^k(x)\right). \end{aligned}$$

Next, we note that the sets $B_n, T(B_n), \ldots, T^{n-1}(B_n)$ are pairwise disjoint. This follows from the fact that if $T^i(B_n) \cap T^j(B_n) \neq \emptyset$ for some $i < j$, then $B_n \cap T^{j-i}(B_n) \neq \emptyset$, but we have already established the family of sets B_i are disjoint, while $T^{j-i}(B_n) \subseteq B_1 \cup \cdots \cup B_{n-(j-i)}$.

We will define the decomposition of $B_1 \cup \cdots \cup B_n$ as follows. Take

$$\begin{aligned} B'_n &= B_n \\ C_n &= B'_n \cup T(B'_n) \cup \cdots \cup T^{n-1}(B'_n) \\ B'_{n-1} &= B_{n-1} \setminus C_n \\ C_{n-1} &= B'_{n-1} \cup T(B'_{n-1}) \cup \cdots \cup T^{n-2}(B'_{n-1}) \\ &\vdots \\ B'_1 &= B_1 \setminus (C_2 \cup \cdots \cup C_n) \\ C_1 &= B'_1. \end{aligned}$$

The picture one must have in mind is viewing each C_k as a "column" in a table with disjoint "levels" given by $T^j\left(B'_k\right)$ for each $0 \leq j \leq k-1$. The "bases" B'_k are pairwise disjoint, as well as the "levels" $T^j\left(B'_k\right)$ for each $0 \leq j \leq k-1$. The union of the columns C_k is equal to $B_1 \cup \cdots \cup B_n$. Furthermore, we note that the bases B'_k are disjoint from the columns C_j for each $1 \leq j \leq k-1$. Therefore, if two columns were to intersect, we would apply T^{-1} until one column intersects the base of the other, which is not the case.

Therefore, we have

$$\begin{aligned} \int_{B_1 \cup \cdots \cup B_n} f \, d\mu &= \sum_{k=1}^{n} \int_{C_k} f \, d\mu \\ &= \sum_{k=1}^{n} \int_{B'_k \cup \cdots \cup T^{k-1}\left(B'_k\right)} f \, d\mu \\ &= \sum_{k=1}^{n} \int_{B'_k} \left(f + f \circ T + \cdots + f \circ T^{k-1}\right) d\mu \\ &\geq 0, \end{aligned}$$

where we use the fact that $f + f \circ T + \cdots + f \circ T^{k-1} > 0$ on $B_k \supseteq B'_k$. □

Corollary: For each $\alpha \in \mathbb{R}$, one has

$$\int_{\{x : f^*(x) > \alpha\}} f \, d\mu \geq \alpha \mu(\{x : f^*(x) > \alpha\}).$$

Proof. Let $g = f - \alpha$. Then, $\{x : f^*(x) > \alpha\} = \{x : g^*(x) > 0\}$, so that

$$\begin{aligned} 0 &\leq \int_{\{x:g^*(x)>0\}} g \, d\mu \\ &= \int_{\{x:f^*(x)>\alpha\}} f - \alpha \, d\mu. \end{aligned}$$

□

We may now prove the pointwise ergodic theorem.

Theorem (Pointwise Ergodic Theorem)**:** Let $(X, \mathcal{B}, \mu)$ be a probability space, and let $T : X \to X$ a measure-preserving transformation. Let $f \in L^1(\mu)$. Then, the following hold:

(i) the function $\overline{f} = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} f \circ T^k(x)$ is defined almost everywhere;

(ii) $\overline{f} \circ T = \overline{f}$ almost everywhere;

(iii) $\overline{f} \in L^1(\mu)$ with $\left\|\overline{f}\right\|_{L^1} \leq \|f\|_{L^1}$;

(iv) if $A \in \mathcal{B}$ is a T-invariant subset, then $\int_A \overline{f} \, d\mu = \int_A f \, d\mu$;

(v) one has $\left(\frac{1}{n} \sum_{k=0}^{n-1} f \circ T^k\right) \to \overline{f}$ in L^1.

Remark: Part (iv) effectively says that $\overline{f} = E[f|\mathcal{I}]$ almost everywhere, where $E[f : \mathcal{I}]$ denotes the conditional expectation onto the σ-algebra of T-invariant subsets.

If X has infinite measure, then (i)–(iii) hold, and (iv) and (v) only hold for finite measure invariant subsets.

Proof.

(i) For each α, β with $\alpha < \beta$, let

$$E_{\alpha,\beta} = \left\{ x \in X : \liminf_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} f \circ T^k(x) < \alpha < \beta < \limsup_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} f \circ T^k(x) \right\}.$$

We will show that $\mu\left(E_{\alpha,\beta}\right) = 0$ for each α, β. It would follow that the union over all rational α, β would have measure 0, so the limit would exist almost everywhere.

Notice that $E_{\alpha,\beta}$ is an invariant subset of $\{x : f^*(x) > \beta\}$. Restricting T to $E_{\alpha,\beta}$, the maximal ergodic theorem implies that

$$\int_{E_{\alpha,\beta}} f \, d\mu \geq \beta\mu\left(E_{\alpha,\beta}\right).$$

We notice that if $x \in E_{\alpha,\beta}$, then there is $n \geq 1$ such that

$$\frac{1}{n} \sum_{k=0}^{n-1} f \circ T^k(x) < \alpha,$$

so that $E_{\alpha,\beta} \subseteq \{x : (-f)(x) > -\alpha\}$. The maximal ergodic theorem then gives

$$\int_{E_{\alpha,\beta}} -f \, d\mu \geq -\alpha\mu\left(E_{\alpha,\beta}\right),$$

or

$$\int_{E_{\alpha,\beta}} f \, d\mu \leq \alpha\mu\left(E_{\alpha,\beta}\right).$$

Therefore, we obtain that $\beta\mu\left(E_{\alpha,\beta}\right) \leq \alpha\mu\left(E_{\alpha,\beta}\right)$, which can only hold when $\mu\left(E_{\alpha,\beta}\right) = 0$.

(ii) This is immediate from the definition of $\overline{f}$.

(iii) By Fatou's Lemma, we have

$$\left|\frac{1}{n}\sum_{k=0}^{n-1} f\circ T^k\right| \leq \frac{1}{n}\sum_{k=0}^{n-1}|f|\circ T^k,$$

so that $\left|\overline{f}\right| \leq \overline{|f|}$, whence

$$\begin{aligned}\int \left|\overline{f}\right| d\mu &\leq \int \overline{|f|}\, d\mu \\ &\leq \liminf_{n\to\infty} \int \frac{1}{n}\sum_{k=0}^{n-1}\left|f\circ T^k\right| d\mu \\ &= \int |f|\, d\mu \\ &< \infty,\end{aligned}$$

where we use the fact that T is measure-preserving in the second-to-last line.

(iv) Suppose A is a T-invariant set. For each $n \geq 0$ and each $k \in \mathbb{Z}$, we let

$$A_{n,k} = \left\{x \in A : \frac{k}{2^n} \leq \overline{f}(x) \leq \frac{k+1}{2^n}\right\}.$$

We have that $A_{n,k}$ are pairwise disjoint invariant sets in k with $A = \bigcup_k A_{n,k}$. Fix $\varepsilon > 0$. Then, $f^* > \frac{k}{2^n} - \varepsilon$ on $A_{n,k}$, meaning

$$\int_{A_{n,k}} f\, d\mu \geq \left(\frac{k}{2^n} - \varepsilon\right)\mu(A_{n,k}).$$

Similarly, $(-f)^* > -\frac{k+1}{2^n}$ on $A_{n,k}$, so that

$$\int_{A_{n,k}} -f\, d\mu \geq -\frac{k+1}{2^n}\mu(A_{n,k}).$$

Letting $\varepsilon \to 0$ gives

$$\frac{k}{2^n}\mu(A_{n,k}) \leq \int_{A_{n,k}} f\, d\mu \leq \frac{k+1}{2^n}\mu(A_{n,k}).$$

Since by definition, we have

$$\frac{k}{2^n}\mu(A_{n,k}) \leq \int_{A_{n,k}} \overline{f}\, d\mu \leq \frac{k+1}{2^n}\mu(A_{n,k}),$$

it follows that

$$\left|\int_{A_{n,k}} f\, d\mu - \int_{A_{n,k}} \overline{f}\, d\mu\right| \leq \frac{1}{2^n}\mu(A_{n,k}).$$

Summing over k gives

$$\left|\int_{A} f\, d\mu - \int_{A} \overline{f}\, d\mu\right| \leq \frac{1}{2^n}\mu(A).$$

Choosing n sufficiently large gives the desired result.

(v) Considering bounded functions yields the desired result through dominated convergence. Else, we approximate by bounded functions, and assume without loss of generality that $f \geq 0$. If $0 \leq g \leq f$ is a bounded function, then

$$\begin{aligned}\left\|\frac{1}{n}\sum_{k=0}^{n-1} f\circ T^k - \overline{f}\right\|_{L^1} &\leq \left\|\frac{1}{n}\sum_{k=0}^{n-1}\left(f\circ T^k - g\circ T^k\right)\right\|_{L^1} + \left\|\frac{1}{n}\sum_{k=0}^{n-1} g\circ T^k - \overline{g}\right\|_{L^1} + \left\|\overline{g}-\overline{f}\right\|_{L^1} \\ &\leq \left\|\frac{1}{n}\sum_{k=0}^{n-1}\left((f-g)\circ T^k\right)\right\|_{L^1} + \left\|\frac{1}{n}\sum_{k=0}^{n-1} g\circ T^k - \overline{g}\right\|_{L^1} + \|f-g\|_{L^1}\end{aligned}$$

Choosing a satisfactory bounded function g gives our desired result.

□

Ergodicity

Definition: We say a measure-preserving map $T: X \to X$ is *ergodic* if, for any $A \subseteq X$, one has $\mu\left(T^{-1}(A)\triangle A\right) = 0$ implies $\mu(A) = 0$ or $\mu(A^c) = 0$.

If μ is a probability measure, then the pointwise ergodic theorem and the ergodicity of the transformation T gives that

$$\frac{1}{n}\sum_{k=0}^{n-1} f\circ T^k(x) \xrightarrow{n\to\infty} \int_X f\, d\mu.$$

Proposition: Let (X,μ) be a probability measure space, and let $T: X \to X$ be a measure-preserving transformation. Then, the following are equivalent:

(i) T is ergodic;

(ii) for all measurable $f: X \to \mathbb{C}$, if $f \circ T = f$ almost everywhere, then there is $\lambda \in \mathbb{C}$ such that $f = \lambda$ almost everywhere (i.e., if f is T-invariant, then f is almost everywhere constant);

(iii) there exists $1 \leq p \leq \infty$ such that if $f \in L^p(\mu)$ is T-invariant, then f is almost everywhere constant;

(iv) there is a total subset $D \subseteq L^2(\mu)$ such that

$$\lim_{n\to\infty}\left\|\frac{1}{n}\sum_{k=0}^{n-1} f\circ T^k - \int_X f\, d\mu\right\|_{L^2} = 0.$$

Proof. The direction (ii) implies (i) follows immediately from the measurable functions χ_A.

To show (i) implies (ii), we see that $\mathrm{Re}(f)$ and $\mathrm{Im}(f)$ are T-invariant, so without loss of generality, f is $\mathbb{R}$-valued. Similarly, f_+ and f_- are T-invariant if f is T-invariant.

Let

$$E_t = \{x \in X : f(x) > t\}.$$

Then,

$$\mu\left(T^{-1}(E_t)\triangle E_t\right) = 0$$

almost everywhere, so by (i), it follows that $\mu(E_t) = 0$ or $\mu(E_t^c) = 0$. If

$$s = \sup\{t \in \mathbb{R} : \mu(E_t^c) = 0\},$$

then

$$\mu(\{x : f(x) \geq s\}^c) = 0.$$

Furthermore, for all $t > s$, we have $\mu(\{x : f(x) > t\}) = 0$, so that

$$\mu(\{x : f(x) < s\}^c) = 0.$$

Thus $f = s$ almost everywhere.

Next, we have that (ii) implies (iii) immediately. Similarly, (iii) implies (i) since $\chi_E \in L^p(X, \mu)$ for all measurable $E \subseteq X$ as μ is a probability measure.

We show now that (ii) implies (iv). By the pointwise ergodic theorem, we have that

$$\overline{f}(x) = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} f \circ T^k(x)$$

is defined almost everywhere, and by the mean ergodic theorem, the latter sum converges in L^2. Passing to a subsequence, we have that

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} f \circ T^k \xrightarrow{L^2} \overline{f}.$$

Since $\overline{f} = \lambda$ by the assumption in (ii), it follows from the pointwise ergodic theorem that

$$\begin{aligned} \lambda &= \int \overline{f}\, d\mu \\ &= \int f\, d\mu. \end{aligned}$$

In particular, this gives

$$\lim_{n\to\infty} \left\| \frac{1}{n} \sum_{k=0}^{n-1} f \circ T^k - \int f\, d\mu \right\|_{L^2} = 0.$$

Next, we have that (iv) implies (iii) by taking $p = 2$. Then, we have

$$\overline{f} = \lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} f \circ T^j$$

has $\int \overline{f}\, d\mu = \int f\, d\mu$ by the pointwise ergodic theorem, and one has

$$\begin{aligned} \left\| \overline{f} - \int f\, d\mu \right\|_{L^2}^2 &= \left\| \overline{f} \right\|_{L^2}^2 - \left| \int f\, d\mu \right|^2 \\ &= \lim_{n\to\infty} \frac{1}{n} \left\| \sum_{j=0}^{n-1} f \circ T^j \right\|_{L^2}^2 - \left| \int f\, d\mu \right|^2. \end{aligned}$$

□

Remark: Letting

$$L_0^2(X, \mu) = \left\{ f \in L^2(X, \mu) : \int f\, d\mu = 0 \right\},$$

then it follows that T is ergodic if and only if for all $f \in L^2_0(X,\mu)$, we have

$$\frac{1}{n}\left\|\sum_{j=0}^{n-1} f\circ T^j\right\|_{L^2} \to 0.$$

Similarly, T is ergodic if and only if the above holds on a total subset of $L^2_0(X,\mu)$. We will occasionally call $L^2_0(X,\mu)$ the space of centered L^2 functions on (X,μ).

Example: Let $T_\alpha : \mathbb{R}/\mathbb{Z} \to \mathbb{R}/\mathbb{Z}$ take $T_\alpha(x+\mathbb{Z}) = (x+\alpha)+\mathbb{Z}$ for some irrational $\alpha \in \mathbb{R}\setminus\mathbb{Q}$. Let $e_n \in \mathbb{C}(\mathbb{R}/\mathbb{Z})$ be given by $e_n(x+\mathbb{Z}) = e^{2\pi i n x}$.

Then, by Stone–Weierstrass, we have that $\overline{\text{span}}^{\|\cdot\|_u}(\{e_n : n\in\mathbb{Z}\}) = C(\mathbb{R}/\mathbb{Z})$ with the supremum norm. Therefore, we have

$$L^2(X,\mu) = \overline{\text{span}}^{\|\cdot\|_{L^2}}(\{e_n : n\in\mathbb{Z}\})$$
$$L^2_0(X,\mu) = \overline{\text{span}}^{\|\cdot\|_{L^2}}(\{e_n : n\in\mathbb{Z}\setminus\{0\}\}).$$

We will see that T_α is ergodic as follows. It suffices to show that for each e_n, we have

$$\begin{aligned}\frac{1}{K}\sum_{j=0}^{K-1} e_n\circ T_\alpha^j(x) &= \frac{1}{K}\sum_{j=0}^{K-1}\sum_{j=0}^{K-1} e^{2\pi i n(x+j\alpha)}\\ &= \frac{e^{2\pi i n x}}{K}\sum_{j=0}^{K-1} e^{2\pi i j n\alpha}\\ &= \frac{e^{2\pi i n x}}{K}\frac{1-e^{2\pi i n K\alpha}}{1-e^{2\pi i n\alpha}}.\end{aligned}$$

Since $\alpha\notin\mathbb{Z}$, it follows that the right-hand side tends to 0 as $K\to\infty$, so by DCT,

$$\left\|\frac{1}{K}\sum_{j=0}^{K-1} e_n\circ T_\alpha^j(x)\right\|_{L^2} \to 0.$$

In particular, we have for all $f\in C(\mathbb{R}\setminus\mathbb{Z})$, we have

$$\frac{1}{K}\sum_{j=0}^{K-1} f\circ T_\alpha^j(x) \to \int_X f\,d\mu.$$

We will prove the normal number theorem, which shows the normal numbers in any subinterval of $[0,1)$ have full measure.

Let $a<b$ be in $[0,1)$. Choose $f\le\chi_{[a,b]}\le g$ with $f,g\in C(\mathbb{R}/\mathbb{Z})$ satisfying $f\ge\chi_{[a+1/n,b-1/n)}$ and $g\le\chi_{[a-1/n,b+1/n]}$. Then,

$$\begin{aligned}b-a-\frac{2}{n} &\le \int f\,dx\\ &= \lim_{K\to\infty}\frac{1}{K}\sum_{j=0}^{K-1} f\circ T_\alpha^j(x)\\ &\le \liminf_{K\to\infty}\frac{1}{K}\sum_{j=0}^{K-1}\chi_{[a,b]}\circ T_\alpha^j(x)\end{aligned}$$

$$\leq \limsup_{K\to\infty} \frac{1}{K} \sum_{j=0}^{K-1} \chi_{[a,b]} \circ T_\alpha^j(x)$$
$$\leq \int g(x)\,dx$$
$$\leq b - a + \frac{2}{n}.$$

It follows then that

$$\frac{|\{0 \leq j \leq K-1 : j\alpha \in [a,b] \bmod 1\}|}{K} \to b - a.$$

Therefore, irrational rotation is ergodic.

Definition: Let T be a probability measure-preserving action on (X, μ) and let S be a probability measure-preserving action on (Y, ν). We say T and S are *isomorphic mod null sets* if there exist null subsets $N \subseteq X$ and $M \subseteq Y$ and a Borel bijection $\varphi\colon X \setminus N \to Y \setminus M$ such that $\varphi \circ T = S \circ \varphi$ almost everywhere.

If φ is not bijective (but is Borel), we call φ a *factor map*.

The notion of isomorphism mod null sets is useful in the following example, where we prove that almost every number in $[0,1]$ normal.

Example: Consider the map $T_n\colon [0,1) \to [0,1)$ given by $T_n(x) = nx \bmod 1$. We consider the map $\varphi\colon [n]^{\mathbb{N}} \to [0,1)$ given by

$$\varphi(x) = \sum_{j=1}^{\infty} \frac{x_j}{n^j}.$$

This is an isomorphism modulo null sets (as the set of numbers in $[0,1)$ with non-unique n-ary expansion is countable), and we observe that

$$\frac{\ell}{n} \leq \varphi(x) < \frac{\ell+1}{n}$$

if and only if $x_1 = \ell$. Notice then that

$$\begin{aligned} T_n(\varphi(x)) &= n\varphi(x) - \ell \\ &= \left(\sum_{j=2}^{\infty} \frac{x_j}{n^{j-1}}\right) + \ell - \ell \\ &= \varphi(T_n(x)). \end{aligned}$$

Therefore, φ gives an isomorphism modulo null sets with the measure-preserving system $([n], u_n)^{\mathbb{N}}$ equipped with the shift action. Since shifts are ergodic on $([n], u_n)^{\mathbb{N}}$, it follows that T_n is ergodic as well.

Let $\ell = 0, \ldots, n-1$, and define

$$I_\ell = \left\{ x \in [0,1) : \frac{\ell}{n} \leq x < \frac{\ell+1}{n} \right\}.$$

Writing the n-ary expansion for x as $x = \sum_{j=1}^{\infty} \frac{x_j}{n^j}$, we have

$$\begin{aligned} \frac{1}{k}\left|\left\{0 \leq j \leq k-1 : x_j = \ell\right\}\right| &= \frac{1}{k}\left|\left\{0 \leq j \leq k-1 : T_n^j(x) \in I_\ell\right\}\right| \\ &= \frac{1}{k} \sum_{j=0}^{k-1} \chi_{I_\ell}\left(T_n^j(x)\right) \end{aligned}$$

$$\to \frac{1}{n}$$

for almost every x, following from the pointwise ergodic theorem.

If we let

$$E_n = \left\{ x \in [0,1) : \lim_{k\to\infty} \frac{1}{k} |\{0 \le j \le k-1 : \text{the } j\text{th digit in the } n\text{-ary expansion is } \ell\}| = \frac{1}{n} \text{ for all } 0 \le \ell \le n-1 \right\}$$

(i.e., E_n is the set of all n-normal numbers), then $E = \bigcap_{n=1}^{\infty} E_n$ is the set of all normal numbers. Since each E_n has measure 1, it follows that E is the intersection of a countable intersection of a family of measure 1 subsets, so it is measure 1.

Mixing, Weak Mixing, and Compactness

For this section, we will concern ourselves with probability measure-preserving actions of either semigroups or groups. The corresponding representations will be either by isometries (in the case of semigroups) or unitaries (in the case of groups).

Definition: A representation $G \to B(H)$ is called mixing if G is infinite and, for all $\xi, \eta \in H$, we have $\lim_{s\to\infty} \langle \pi(s)\xi, \eta\rangle = 0$. That is, for all $\varepsilon > 0$, there exists some finite subset $F \subseteq G$ such that for all $s \in G\backslash F$, we have $\left|\langle \pi(s)\xi, \eta\rangle\right| < \varepsilon$.

The representation is called weak mixing if, for all $\xi_1, \dots, \xi_k \in H$ and all $\varepsilon > 0$, there exists $s \in G$ such that $\left|\left\langle \pi(s)\xi_i, \xi_j \right\rangle\right| < \varepsilon$ for all $i, j = 1, \dots, k$.

We recall the definition of ergodic representations here.

Definition: A representation $G \to B(H)$ is ergodic if G is infinite and $\pi(s)\xi = 0$ for all $s \in G$ implies $\xi = 0$.

For both of these definitions, it suffices to verify on a total subset.

The following two facts are very useful.

- We have the isomorphism $L^2(X,\mu) \otimes L^2(Y,\nu) \cong L^2(X \times Y, \mu \otimes \nu)$.
- If $S \curvearrowright (X,\mu)$ is a semigroup acting by probability measure-preserving transformations, then the Koopman representation $\alpha\colon S^{\text{op}} \to B\left(L_0^2(X)\right)$ is given by $\alpha_{s^{\text{op}}} f(x) = f(s.x)$.

Proposition: Let $S \curvearrowright (X,\mu)$ be a semigroup acting by probability measure-preserving transformations, and let $\alpha\colon S^{\text{op}} \to B\left(L_0^2(X)\right)$ be the corresponding Koopman representation. Then, the following are equivalent:

(i) α is weak mixing;

(ii) for all probability measure-preserving actions $S \curvearrowright (Y,\nu)$ with corresponding Koopman representation $\beta\colon S^{\text{op}} \to B\left(L_0^2(Y)\right)$, $\alpha \otimes \beta$ has no nonzero invariant vectors;

(iii) $\alpha \otimes \alpha$ has no nonzero invariant vectors.

Proof. We start by showing that (i) implies (iii). Consider the total subset

$$D = \left\{ f \otimes g : f \in L_0^2(X), g \in K \right\},$$

where K is the maximal S-invariant subspace of $L_0^2(Y)$. We have for all $s \in S$ that

$$\begin{aligned} \langle (\alpha_{s^{\text{op}}} \otimes \beta_{s^{\text{op}}})(f_1 \otimes g_1), f_2 \otimes g_2 \rangle &= \langle \alpha_{s^{\text{op}}}(f_1), f_2 \rangle \langle \beta_{s^{\text{op}}}(g_1), g_2 \rangle \\ &= \langle \alpha_{s^{\text{op}}}(f_1), f_2 \rangle \langle g_1, g_2 \rangle. \end{aligned}$$

We observe that $\alpha_{s^{\text{op}}}(f_1) = f_1$ if and only if $f_1 = 0$, so that we have that $f_1 \otimes g_1 = 0$.

Next, we show that (iii) implies (i). Suppose not. Then, there exists some $\varepsilon > 0$ and some $\xi_1, \dots, \xi_k \in L^2_0(X)$ such that, for all $s \in S$,

$$\sum_{i,j=1}^{k} \left| \left\langle \pi_s \xi_i, \xi_j \right\rangle \right|^2 \geq \varepsilon^2.$$

Let

$$\xi = \sum_{i=1}^{k} \xi_i \otimes \overline{\xi_i}.$$

Let

$$C = \overline{\text{conv}}\{(\alpha_{s^{\text{op}}} \otimes \alpha_{s^{\text{op}}})\xi : s \in S\}.$$

By the Hilbert projection theorem, there is some $\eta \in C$ with minimal norm. We claim that $\eta \neq 0$. This follows from the fact that

$$\begin{aligned}
\text{Re}(\langle (\alpha_{s^{\text{op}}} \otimes \alpha_{s^{\text{op}}})\xi, \xi \rangle) &= \text{Re}\left(\sum_{i,j=1}^{k} \left\langle \alpha_{s^{\text{op}}} \xi_i \otimes \alpha_{s^{\text{op}}} \overline{\xi_i}, \xi_j \otimes \overline{\xi_j} \right\rangle \right) \\
&= \text{Re}\left(\sum_{i,j=1}^{k} \left\langle \alpha_{s^{\text{op}}} \xi_i, \xi_j \right\rangle \left\langle \alpha_{s^{\text{op}}} \overline{\xi_i}, \overline{\xi_j} \right\rangle \right) \\
&= \sum_{i,j=}^{k} \left| \left\langle \alpha_{s^{\text{op}}} \xi_i, \xi_j \right\rangle \right|^2 \\
&\geq \varepsilon^2.
\end{aligned}$$

Thus, $\text{Re}(\langle \eta, \xi \rangle) \geq \varepsilon^2 > 0$, so that $\eta \neq 0$. Since S acts by probability measure-preserving transformations, it follows that

$$\left\| (\alpha_{s^{\text{op}}} \otimes \alpha_{s^{\text{op}}})\eta \right\| = \left\| \eta \right\|$$

with $(\alpha_{s^{\text{op}}} \otimes \alpha_{s^{\text{op}}})\eta \in C$ for all $s \in S$, so that η is fixed by S. □

Definition: A probability measure-preserving transformation $S \curvearrowright (X, \mu)$ is called weak mixing if for all $\varepsilon > 0$ and all measurable $A_1, \dots, A_k \subseteq X$, there exists some $s \in S$ such that

$$\left| \mu\left(s^{-1}.A_i \cap A_j\right) - \mu(A_i)\mu\left(A_j\right) \right| < \varepsilon.$$

Theorem: Let $S \curvearrowright (X, \mu)$ be a probability measure-preserving transformation. The following are equivalent:

(i) the action $S \curvearrowright (X, \mu)$ is weak mixing;

(ii) $\alpha \colon S^{\text{op}} \to L^2_0(X, \mu)$ is weak mixing;

(iii) the diagonal action $S \curvearrowright (X \times X, \mu \otimes \mu)$ is ergodic.

Proof. □